

No. of Functional Features Included in the Solution

Date	15 July 2026
Team ID	SWTID-2026-4513
Project Name	Credit card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing	Cleans dataset, handles missing values, performs feature engineering and encoding.	Data Processing	Done	High
2	Credit Card Approval Prediction	Predicts whether a customer is approved or rejected using the trained Random Forest model.	Machine Learning Model	Done	High
3	Applicant Data Input Form	Collects applicant details through a user-friendly web interface.	Frontend (Flask Templates)	Done	Medium
4	Prediction Result Display	Displays approval/rejection status based on model prediction.	Flask Backend & UI	Done	High
5	Model Performance Evaluation	Evaluates the model using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix.	Model Evaluation	Done	Medium
6	Feature Importance Analysis	Shows the importance of input features used by the Random Forest model.	Machine Learning	Done	Medium
7	Responsive Web Interface	Provides a responsive UI using Bootstrap for better user experience.	Frontend	Done	Medium
8	Project Deployment	Deploy the Flask application on a cloud platform (Render).	Deployment	In Progress (or Done if your deployment succeeds)	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	Applicant data entry and result display	Prediction Result Page, Responsive Bootstrap Interface
2	Backend / Logic	Data processing, feature engineering, and prediction logic	Data Cleaning, Label Encoding, Random Forest Prediction, Flask Backend
3	Database / Storage	Model storage and input data handling	credit_card_model.pkl, CSV Dataset (application_record.csv, credit_record.csv)
4	API / Integration	Flask routing and HTTP request handling	Flask Routes (/ , /predict, /result), Form Submission using POST
5	Security / Authentication	Input validation and exception handling	Server-side Input Validation, Try-Except Error Handling (<i>No user authentication implemented</i>)